Deep Learning-Based Intelligent Visual Diagnosis of Tomato Yellow Leaf Curl Virus Using Leaf Images: A Review

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***Abstract*—Tomato yellow leaf curl virus (TYLCV) is one of the most damaging pathogens of tomato, and early, accurate field diagnosis is a prerequisite for timely vector control and removal of infected plants. Because its symptoms are visible on leaves, deep-learning analysis of leaf images is an attractive screening route, and the plant-disease imaging literature has grown rapidly. This review synthesises that literature from the perspective of TYLCV. We cover the virus and its diagnostic problem, imaging modalities, deep-learning architectures (convolutional networks, transformers, detectors, segmentation and vision-language models), training strategies, datasets, evaluation metrics, calibration, uncertainty and explainability, and deployment on mobile and edge devices. We compare model families and study designs and map the strength of evidence by theme. Three findings recur: near-perfect accuracy on lab-background benchmarks does not transfer to field images; benchmark backgrounds and near-duplicate images can inflate results through shortcuts and leakage; and labels are rarely confirmed by molecular tests. We found no field-validated TYLCV study with molecularly confirmed ground truth. We identify gaps, propose a six-level evaluation maturity ladder and a reporting checklist, and outline research directions for field-ready TYLCV leaf diagnosis.**

***Index Terms*—tomato yellow leaf curl virus, deep learning, convolutional neural networks, vision transformers, transfer learning, plant disease diagnosis, explainable AI, uncertainty, domain shift, precision agriculture.**

I. INTRODUCTION

Tomato is one of the most widely grown vegetable crops, and tomato yellow leaf curl virus (TYLCV) is among its most damaging pathogens, capable of causing near-total loss of marketable fruit when plants are infected early [1]. The virus is transmitted by whiteflies in a persistent manner [2], can also be seed-transmissible [3], and has spread worldwide through trade and vector movement [4]. No curative treatment exists for an infected plant, so control depends on keeping vectors away, growing resistant cultivars and removing infected plants early [1], [5], [6]. Early removal presupposes early detection, and in practice detection relies on trained observers recognising upward leaf curling, marginal yellowing and stunting, a skill that is scarce and subjective at the scale of commercial production.

Deep learning has changed image-based plant disease diagnosis. Convolutional neural networks (CNNs) trained on leaf photographs reached accuracies above 99% on the PlantVillage benchmark [7], [8], and a large number of surveys now cover the field [9], [10], [11], [12], [13], [14], [15], [16], [17], [18]. Yet the earliest large study already reported a sharp drop when models were tested on images taken under different conditions [7], and later work argues that benchmark performance overstates what can be expected in the field [19], [20], [21], [22]. For TYLCV, whose management depends on rapid action, an overstated accuracy has real costs.

A. Contributions of This Review

This paper is a narrative review that asks how far deep-learning leaf-image diagnosis can currently be trusted for TYLCV. Its contributions are: (i) a TYLCV-centred synthesis of the plant-disease imaging literature, spanning architectures, training, datasets, metrics, calibration, uncertainty, explainability and deployment; (ii) a comparison of model families, study designs and evidence strength by theme; (iii) an explicit list of research gaps; and (iv) a six-level evaluation maturity ladder and a reporting checklist for future TYLCV studies.

B. Review Methodology

We searched web and library sources for work on image-based plant disease diagnosis, tomato leaf disease classification, dataset bias and generalisation, detection, segmentation and severity estimation, lightweight deployment, calibration, uncertainty and explainability, and on TYLCV biology and management. The scope is roughly

2016–2026, with a small number of foundational deep-learning papers retained where they define the methods discussed. Studies were selected for relevance to leaf-image diagnosis and for influence in the field, then grouped by theme and compared qualitatively because they differ in datasets, architectures and evaluation, which prevents formal meta-analysis. This is not a systematic review: no protocol was registered, the search was not exhaustive, and numerical results are quoted only where we are confident of them.

C. Relation to Existing Surveys

Table I positions this review against recent surveys. Existing surveys cover plant disease imaging broadly and describe architectures, datasets and open problems; to our knowledge none of those listed focuses on a single virus such as TYLCV or on the evidence needed for field use of a screening tool.

TABLE I
COMPARISON WITH EXISTING SURVEYS

Survey	Scope	Emphasis
[9]	Deep learning in agriculture	Broad agricultural applications
[10], [11], [13]	Plant disease deep learning	Architectures, datasets, comparisons
[12], [16]	Image processing / vision-based ML	Pipelines and systematic trends
[14], [15], [17]	Disease and pest detection with DL	Detection, CNN reviews, recent advances
[18]	Diagnosis tools and recommendations	Deployment and tool development
[23], [24]	IoT, data fusion, remote sensing	Multi-source and large-scale monitoring
This review	TYLCV-centred, field-evidence focus	Evidence strength, evaluation ladder, checklist

The remainder of the paper is organised as follows. Section II gives background on TYLCV, sensing and deep learning. Section III reviews the literature. Section IV compares the evidence, Section V lists research gaps, Section VI discusses challenges, Section VII proposes directions and a roadmap, and Section VIII concludes.

II. BACKGROUND AND RELATED WORK

A. TYLCV Biology and Epidemiology

TYLCV is a monopartite begomovirus (family *Geminiviridae*) with a circular single-stranded DNA genome of roughly 2.8 kb [1]. It is transmitted by the whitefly *Bemisia tabaci*, in which virions circulate through the gut and haemolymph to the salivary glands before inoculation [2]. Its worldwide spread and recombination with related begomoviruses make it an evolving target [4], [25]. Resistance genes introgressed from wild tomato relatives are widely used but are not fully durable [5], and whitefly control is complicated by insecticide resistance [26] and by changing climate [27]. Integrated programmes therefore combine exclusion, resistant cultivars, vector monitoring and early removal of infected plants [6], which is where leaf-image screening would fit.

B. Symptoms and the Diagnostic Problem

Infected tomato plants show leaflets that curl upward and inward, yellowing at leaf margins, shortened internodes, a dwarfed bushy habit and reduced flowering and fruit set; severity depends on the plant's age at infection [1]. These cues are visible on leaves, but similar symptoms can arise from other begomoviruses, nutrient disorders, herbicide injury or heat stress, so visual diagnosis is a screening step and molecular tests (PCR/qPCR) remain the confirmatory standard [1]. A useful image-based system therefore has to separate TYLCV from look-alike conditions and from healthy leaves, not merely recognise yellow leaves.

C. Sensing Modalities

RGB imaging with consumer cameras is the cheapest and most scalable modality. Hyperspectral and thermal sensors can detect physiological changes before visible symptoms appear, but cost and calibration demands limit them to research and phenotyping settings [28]. Remote sensing from drones and satellites supports monitoring at field and regional scales [24], and surveys that fuse imaging with IoT sensors and environmental data point to multi-modal systems as a longer-term direction [23]. Table II summarises the trade-offs.

TABLE II
IMAGING MODALITIES FOR LEAF DISEASE DIAGNOSIS

Modality	Strengths	Limitations
RGB (phone)	Cheap, ubiquitous, field-ready	Symptom-stage only; lighting and background vary
Hyperspectral / thermal	Pre-symptomatic signals [28]	Cost, calibration, throughput
Remote sensing [24]	Field- to region-scale monitoring	Coarse resolution; symptom ambiguity
Image + IoT / weather [23]	Context-aware risk and diagnosis	Integration and data-quality burden

D. Deep Learning Building Blocks

Early plant-disease studies fine-tuned general-purpose CNNs on leaf photographs [29], [7]. Widely used CNN families include Inception-style networks [30], residual networks that made very deep models trainable [31], densely connected networks [32] and compound-scaled families [33]; EfficientNet has been applied directly to plant leaf classification [34]. Mobile-oriented designs such as MobileNets [35], MobileNetV2 [36] and MobileNetV3 [37] target on-device inference. Vision transformers treat an image as a sequence of patches [38], and hierarchical

variants such as Swin Transformer scale to dense tasks [39]. For localisation, two-stage and single-stage detectors, exemplified by Faster R-CNN, SSD and YOLO, predict boxes around symptomatic regions [40], [41], [42], and transformer detectors have been proposed as end-to-end alternatives [43]. Semantic segmentation networks [44] and promptable foundation models such as the Segment Anything Model [45] delineate lesions and leaf area, and vision-language models align images with text [46]. Figure 1 shows a reference pipeline.

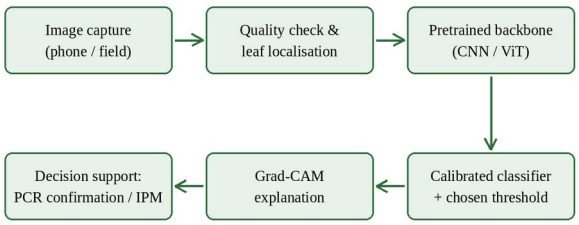


Fig. 1. Reference pipeline for leaf-image TYLCV screening with calibrated outputs and explanations.

E. Training Strategies

Transfer learning from ImageNet is the default because plant-disease datasets are small relative to the models trained on them [47], [19], [48]. Better ImageNet models generally transfer better, although pretraining helps less on some fine-grained tasks [49]. Training practice then determines outcomes. Augmentation (flips, rotations, crops, photometric changes) is universal [50], and learned or mixed augmentations such as AutoAugment and mixup are also used [51], [52]; for TYLCV, photometric changes must be small because yellowing is the diagnostic cue. Class imbalance is handled by class-weighted or re-weighted losses [53], [54]. A staged schedule that trains the classifier head first and then unfreezes the backbone at a lower learning rate, with a decoupled-weight-decay optimiser [55], is a standard way to adapt pretrained networks. Self-supervised pretraining [56], [57], [58] offers a route to use unlabelled images.

F. Evaluation Metrics and Protocols

Accuracy alone is a poor summary for screening: when healthy plants outnumber infected ones, a model that always answers “healthy” scores high accuracy. Sensitivity (recall) is the fraction of infected leaves found, specificity the fraction of healthy or look-alike leaves correctly cleared, and the area under the ROC curve summarises ranking quality across thresholds; for rare positives, precision–recall curves are more informative. Calibration is reported by the expected calibration error [59]. Protocol matters as much as the metric: random image-level splits can put near-identical photographs of the same plant in training and test sets, whereas external testing at another site or with another device measures robustness to distribution shift [60], [61]. Table III lists the main metrics and their pitfalls.

TABLE III
EVALUATION METRICS AND COMMON PITFALLS

Metric	What it tells	Pitfall
Accuracy	Overall fraction correct	Misleading under class imbalance
Sensitivity (recall)	Share of infected leaves found	Depends on the threshold; hides false alarms
Specificity	Share of healthy / look-alike leaves cleared	Meaningless without hard negatives
ROC-AUC / PR-AUC	Ranking quality across thresholds	Not an operating point; PR better for rare positives
ECE / reliability	Reliability of probabilities [59]	Needs enough data per bin
External-set performance	Robustness to shift [60], [61]	Rarely reported

G. Calibration, Uncertainty and Explainability

A diagnostic score is only useful if its probabilities mean what they say. Modern networks tend to be over-confident, and simple post-hoc temperature scaling on a validation set is an effective remedy [59]. Uncertainty can also be estimated with deep ensembles [62] or Monte-Carlo dropout [63], although predictive uncertainty degrades under dataset shift [61]. For explanation, gradient-based saliency such as Grad-CAM highlights the image regions that drove a decision [64] and has been used to check whether tomato disease classifiers attend to lesions [65]; model-agnostic methods such as LIME and SHAP attribute predictions to input features [66], [67]. Explainable pipelines have been used to identify which symptoms drive plant-stress predictions [68], but saliency methods should themselves be validated, because some produce similar maps even for randomised models [69].

III. LITERATURE REVIEW

Figure 2 organises the literature along the axes used in this review: the task, the model family, the data, the measures of trust and the deployment route.

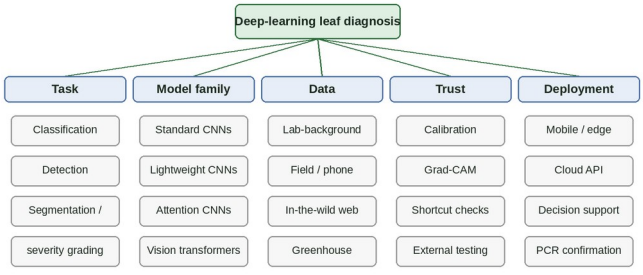


Fig. 2. Taxonomy of deep-learning leaf-image diagnosis used to organise this review.

A. Datasets

PlantVillage is the dominant benchmark: tens of thousands of leaf images across many crop–disease classes photographed in controlled conditions with plain backgrounds [7]. Its convenience produced a large literature but also its main weakness. One analysis showed that PlantVillage backgrounds alone carry enough information to classify images well above chance [22], an instance of the shortcut learning in which networks exploit spurious cues instead of the intended signal [70]. Datasets collected under real conditions are smaller and harder: PlantDoc assembles internet images with annotated bounding boxes for many diseases [71], and cassava studies used field images captured on smartphones [72]. For TYLCV specifically, public sets include a Roboflow Universe dataset [73] and a Kaggle image folder [74]; an inspection of the latter for this review found 100 images, all TYLCV-positive, mostly lab-style photographs. Table IV summarises the main dataset types.

TABLE IV
DATASET TYPES FOR PLANT DISEASE IMAGE ANALYSIS

Dataset type	Characteristics	Main limitation
PlantVillage [7]	About 54k images, 38 classes, plain lab backgrounds	Background shortcuts [22]
PlantDoc [71]	About 2.6k in-the-wild images, annotated boxes	Small, uneven classes
Cassava field images [72]	Smartphone images from real fields	Single crop
Tomato greenhouse images [75]	Real greenhouse scenes, severity labels	Few diseases; limited public access
TYLCV sets [73], [74]	Public TYLCV image collections	Composition and labels unverified; negatives often missing

B. Tomato Leaf Disease Classification

Many studies classify tomato leaf diseases, almost all on PlantVillage tomato classes, with TYLCV as one class among several. Reported work includes comparisons of standard CNNs [76], [65], compact networks for on-device or robotic use [77], lightweight CNNs for modest hardware [78], and attention-augmented networks, including attention-embedded residual networks [79] and squeeze-and-excitation variants [80]. Reported accuracies on this benchmark are uniformly high, which leaves little room to rank methods and little evidence about field behaviour [47], [8]. We did not find, in the sources consulted, a field-validated study dedicated to TYLCV leaf images with molecular ground truth.

C. Lessons from Other Crops

Work on other crops informs what TYLCV studies should expect. Nine-layer and other purpose-built CNNs reached high accuracy on lab images of many species [81]; lightweight CNNs have been applied to rice diseases and pests [82]; and cassava studies relied on field images from real farms [72]. Reviews of CNNs for plant disease conclude that the field conditions of real images, rather than the architecture, are the main obstacle [83], and work addressing current limitations of plant-disease detection stresses dataset size, variety and real-world capture [20].

D. Generalisation to Field Conditions

The gap between laboratory and field is the central finding of the literature. A network that reached very high accuracy on held-out PlantVillage images dropped to roughly 31% on images collected under different conditions [7]. Studies of dataset size and variety conclude that capturing diverse conditions matters more than adding images of the same kind [19], and that classifying individual lesions and spots can be more robust than classifying whole leaves [84]. Work on smartphone capture in the wild shows that training with field-captured images is needed to approach usable field performance [85]. A study re-examining plant-disease characterisation argues for training that reflects practical variation rather than single-source benchmarks [21].

E. Detection, Segmentation and Severity

When several symptoms or plants appear in one frame, detection is more informative than classification. A real-time detector for tomato diseases and pests used region-based detectors with several backbones on field images [86], and a greenhouse system combined detection with severity estimation [75]. General detectors [40], [41], [42], [43] and segmentation models [44], [45] provide the technical basis for lesion localisation and severity grading. Detection requires bounding-box or mask annotation, which is costly, and no TYLCV-specific annotated set of this kind was identified here.

F. Lightweight Models and Deployment

Field use favours models small enough for phones or edge devices [35], [37], [18]. Surveys emphasise practical requirements beyond accuracy: latency, connectivity, user interface and recommendations tied to the diagnosis [18], [23]. Reviews also caution that deployment studies remain scarce relative to benchmark studies [15], [17], [16].

G. Explainability and Trust

Saliency maps were used early to inspect tomato disease classifiers [65] and remain the standard tool for spotting shortcut behaviour: a model that attends to the background rather than the leaf is a warning sign [64], [70]. Saliency is a diagnostic aid, not proof of correctness, and should be

combined with out-of-domain testing and, where possible, attribution methods [66], [67], [69].

H. Look-alike Discrimination

TYLCV is one of several conditions that curl and yellow tomato leaves, and discriminating among them is the clinically relevant task. Multi-class benchmarks include other tomato diseases, but these are mostly fungal or bacterial spots whose appearance differs strongly from a viral curl, so they are easy negatives [65], [76]. Genuinely hard negatives, such as other begomovirus infections, nutrient disorders and herbicide injury, are rare in public data. Detection of subtle or early symptoms may benefit from lesion-level analysis [84] or from non-visible wavelengths [28], but neither has been demonstrated for TYLCV with molecular ground truth.

I. Distribution Shift and Uncertainty

General benchmarks of in-the-wild distribution shift show that models that look excellent in-distribution can fail under shift [60], and that predictive uncertainty degrades as the data move away from the training distribution [61]. These findings match the plant-disease observation of lab-to-field degradation [7] and argue for external test sets, domain-generalisation methods and uncertainty-aware outputs [62], [63] in TYLCV screening.

J. Label-Efficient and Foundation-Model Approaches

Self-supervised methods such as contrastive learning and self-distillation learn visual representations from unlabelled images [56], [57], [58], which could ease the shortage of labelled leaf images, and vision-language models enable classification from text prompts [46]. Promptable segmentation models can assist annotation [45]. We did not find, in the sources consulted, studies applying these approaches specifically to TYLCV leaf images, so their benefit for this virus remains untested.

K. Data Augmentation and Synthetic Data

Augmentation is the standard remedy for small datasets, and surveys catalogue geometric, photometric and generative approaches [50]. For TYLCV, photometric augmentation needs care because yellowing is the diagnostic cue, and synthetic images cannot replace field images with confirmed labels.

L. Reported Accuracy and What It Means

Many studies report accuracies above 95% on tomato leaf classification [8], [47], [79]. Such numbers describe performance on images drawn from the same distribution as the training data, with the same backgrounds and often the same capture sessions. They are informative about model capacity, not about whether a farmer's phone photograph taken in a field will be classified correctly [7], [21]. A fair reading of the literature therefore treats benchmark accuracy as an upper bound rather than an estimate.

IV. COMPARATIVE ANALYSIS OF EXISTING RESEARCH

Table V compares the main model families, Table VI compares study designs and Table VII maps the strength of evidence by theme. The key distinction is not architecture but evaluation: lab-background benchmarks reward capacity and tuning, whereas field-style, duplicate-aware and externally validated evaluations measure what matters for TYLCV screening.

TABLE V
COMPARISON OF MODEL FAMILIES

Model family	Strengths	Limitations for TYLCV
Standard CNNs [7], [31], [30], [32]	Strong, well understood, easy to fine-tune	Large; benchmark-tuned
Lightweight CNNs [33], [35], [37]	Fast, small; mobile-friendly	Capacity trade-offs; little field evidence
Attention CNNs [79], [80]	Small gains on benchmarks	Gains within benchmark saturation
Vision transformers [38], [39]	Strong general vision results	Data-hungry; little plant field evidence
Detectors [40], [41], [42], [43]	Localise several symptoms per image	Costly box annotation; few TYLCV sets
Segmentation / foundation [44], [45], [46]	Lesion delineation; promptable, zero-shot	Untested for TYLCV

TABLE VI
COMPARISON OF STUDY DESIGNS

Study design	Typical evidence	Main limitation
Lab-background classification [7], [8], [47]	Very high benchmark accuracy	Background shortcuts; large drop off-distribution [7], [22]
Attention / lightweight CNNs [79], [80], [78]	Small gains over standard CNNs	Gains within benchmark saturation
Detection / severity [86], [75]	Localises symptoms in real scenes	Costly annotation; few TYLCV sets
Field / smartphone capture [85], [72], [19]	Most realistic accuracy	Smaller datasets; lower accuracy
Explainability checks [64], [65]	Exposes shortcuts	Qualitative; not proof

Architectures themselves matter less than data. Comparative fine-tuning studies find that several mainstream networks reach similar benchmark accuracy [47], while studies of data variety find large effects from the diversity of training images [19]. For TYLCV, the available evidence is therefore dominated by benchmark results that do not isolate the virus from look-alike conditions. A second observation concerns reporting: many papers give a single accuracy on a random split, without confidence intervals, external validation, specificity against look-alikes or calibration. Without these, two methods that differ by one percentage point on a saturated benchmark cannot be

ranked, and neither can be trusted in the field [59], [17]. Reviews consistently call for standardised protocols and more realistic datasets [15], [14], [18], [83].

Table VII summarises how strong the evidence is for each theme, separating what is well established from what remains open.

TABLE VII
EVIDENCE MAP BY THEME

Theme	Strength of evidence	Key sources
Benchmark accuracy	Strong: many studies, consistent results	[7], [8], [47]
Field generalisation	Moderate: mostly other crops	[19], [85], [72], [21]
Dataset bias and shortcuts	Moderate: few dedicated studies	[22], [70]
Detection and severity	Limited for tomato	[86], [75]
Calibration and uncertainty	Strong in general ML; rare in plant studies	[59], [61], [62]
Foundation / self-supervised models	None found for TYLCV	[46], [58], [56]
TYLCV-specific field validation	None found in the sources consulted	—

V. RESEARCH GAPS

1) *Molecularly confirmed labels.* Public image sets are commonly labelled by visual inspection; large-scale TYLCV image sets with PCR-confirmed ground truth are, to our knowledge, not publicly available [1], [74].

2) *Negatives and look-alikes.* Evaluation needs healthy leaves and conditions that mimic TYLCV (other begomoviruses, nutrient and herbicide injury), otherwise specificity is unknown [1], [28].

3) *Field generalisation.* Accuracy collapses when conditions change [7], [19], and backgrounds leak labels [22], [70].

4) *Leakage-aware splitting.* Near-duplicate and same-plant images inflate scores unless splits are group-aware; most reviewed studies do not report such precautions.

5) *Calibration, uncertainty and thresholds.* Most studies report accuracy only, not calibrated probabilities, uncertainty or operating points [59], [62].

6) *Early and pre-symptomatic detection.* RGB models detect visible symptoms; earlier detection needs richer sensing [28], [23], [24].

7) *Foundation and label-efficient models.* Self-supervised and vision-language approaches are untested for TYLCV [46], [58].

8) *Deployment evidence.* Few studies evaluate end-to-end field use including user decisions [18], [15].

VI. CHALLENGES AND LIMITATIONS

Class imbalance and small field datasets limit training. Symptom expression varies with cultivar, plant age, temperature and co-infections, so appearance is not a stable label [1]. Imaging conditions (illumination, camera, leaf pose) differ between users. Saliency maps can be misleading if over-interpreted [64], [69]. Labels derived from visual inspection carry noise, and molecular confirmation at scale is costly. Regulatory and trust issues arise if a screening tool influences decisions about removing plants. Finally, this review is limited by its narrative design, its reliance on published summaries, and the absence of TYLCV-specific field studies to compare.

Ethical and practical considerations also apply. Geo-tagged farm images can reveal field locations and should be handled with consent and appropriate privacy; access to phones, connectivity and extension support differs between growers, so tools should degrade gracefully and be tested with smallholders as well as commercial farms [18]. Automated screening should be presented as decision support with its uncertainty, not as a replacement for expert or laboratory diagnosis, and its limits should be stated to users.

VII. FUTURE RESEARCH DIRECTIONS

We propose a six-level evaluation maturity ladder (Fig. 3). Most published studies sit at the first level; field use of a TYLCV screening tool should rest on evidence from at least levels L2–L4, with L5 and L6 needed before the tool influences management decisions.

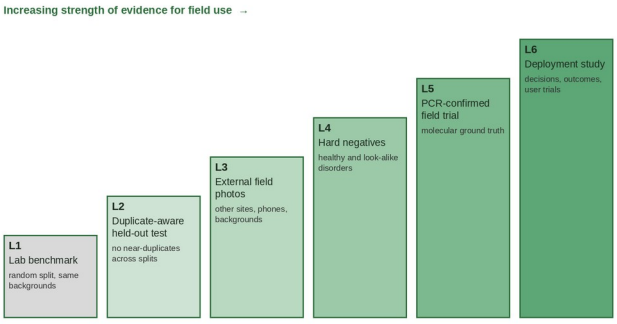


Fig. 3. Evaluation maturity ladder for TYLCV leaf-image diagnosis, from lab benchmark to deployment study.

Table VIII maps the research gaps to concrete directions. In addition, the limitations of image-only diagnosis suggest a complementary design: combine a calibrated image classifier with contextual evidence such as whitefly density, weather and local virus prevalence, which together determine the prior probability of infection [23], [1]. In such a system the image model need not be perfect; a positive screen raises the priority of PCR testing and of vector scouting, and a negative screen in a high-risk context should not suppress monitoring. This framing also clarifies the cost of errors: a false negative delays removal of infected plants

and allows spread, whereas a false positive costs a laboratory test.

TABLE VIII
RESEARCH GAPS AND PROPOSED DIRECTIONS

Gap	Direction
PCR-confirmed labels (1)	Multi-site, multi-device dataset with molecular ground truth
Negatives, look-alikes (2)	Include healthy and hard-negative disorders; report per-condition false positives
Field generalisation (3)	External field sets; background-only and Grad-CAM shortcut checks [64], [22]
Leakage (4)	Group-aware splits; document duplicate handling
Calibration, uncertainty (5)	Temperature scaling, ensembles; thresholds fixed on validation data [59], [62]
Early detection (6)	Hyperspectral or multi-modal studies [28], [23]
Foundation models (7)	Self-supervised pretraining and vision-language tests [56], [58], [46]
Deployment (8)	On-device models and field user studies [37], [18]

Table IX gives a reporting checklist consistent with the ladder, so that readers can judge which level a study has reached.

TABLE IX
PROPOSED REPORTING CHECKLIST

Item	Requirement
Labels	PCR/qPCR-confirmed where possible
Negatives	Healthy plus look-alike disorders
Splits	Group-aware; no near-duplicates across splits
External test	Different sites, cameras and backgrounds
Metrics	Sensitivity, specificity, AUC with CIs; calibration
Shortcut checks	Grad-CAM, background-only baseline
Reporting	Threshold fixed on validation data; code and data availability

A. Recommended Design for a New TYLCV Study

Putting the above together, we recommend the following sequence. (1) Define the task and the negative classes, including look-alike disorders. (2) Collect images across sites, devices and growth stages, with a PCR-confirmed subset. (3) Remove duplicates and assign near-duplicate groups before splitting. (4) Choose a lightweight pretrained backbone [37], [33] and at least one larger baseline [31], [38]. (5) Train with staged fine-tuning, class weighting and small photometric augmentation [47], [53], [50]. (6) Fit temperature scaling and fix the decision threshold on validation data only [59]. (7) Evaluate on the internal test set, an external field set and hard negatives, reporting confidence intervals. (8) Run background-only and Grad-CAM shortcut checks [64], [22]. (9) Report against Table IX and state the level reached on the ladder.

VIII. CONCLUSION

Deep learning can recognise TYLCV-like symptoms in leaf images, but the evidence that it works in the field is weak. High benchmark accuracies are not reliable guides to field behaviour because of lab backgrounds, label shortcuts, duplicate leakage and the absence of molecularly confirmed negatives, and we found no field-validated TYLCV study with PCR ground truth. Progress requires better data and stricter evaluation more than new architectures. With group-aware splits, external field tests, calibrated thresholds, uncertainty estimates and explanation checks, image-based screening can become a trustworthy first step ahead of PCR confirmation and vector management; the maturity ladder and checklist offered here are intended to make that progress measurable.

. ACKNOWLEDGMENT

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. DATA AVAILABILITY

No new datasets were generated for this review. Public datasets discussed are cited in the reference list; their licences should be checked before reuse.

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